# Process & Decision Documentation

## Process Overview Visualization (Group Work Only)

## Project/Assignment Decisions

### Major Decisions

**Decision 1: Converting the single "almost impossible" level into a multi-level difficulty curve**

In A2, the game effectively had one primary level tuned to be extremely hard. Many players in user testing reported that it was nearly impossible to beat and that they struggled to even understand the core mechanic under that pressure. For A3, the team made a foundational decision to add multiple levels and distribute difficulty. The new structure:

- Introduces an easier tutorial/early level where the Puzzle Matching mechanic and pop-ups can be learned without being overwhelmed.
- Gradually increases complexity, with later levels featuring more challenging board layouts (e.g., more complex tile configurations, colours that are harder to distinguish, tighter timing).
- Reserves "unpassable" or interpretive difficulty for the highest tiers, rather than starting there.

This decision addressed accessibility and learning while still supporting the artistic goal of communicating how anxiety can escalate over time.

**Decision 2: Adding a System Stability HUD (health bar) synced with the timer**

To make time pressure and difficulty more legible across levels, the team added a visible stability/health bar at the bottom of the HUD, synchronized with the level timer. Instead of only a number counting down, players now see a bar that drains as time passes and as the level progresses. The bar's pacing was tuned so that its depletion matches the underlying

timer. This bar acts as a visual metaphor for mental stability under GAD: lower levels deplete more gently, while higher levels bring the bar into the critical zone more rapidly.

**Decision 3: Introducing screen shake at critical stability levels on harder levels**

To reflect somatic aspects of anxiety (e.g., trembling, rapid heartbeat), the team implemented a screen shake effect tied to the health bar. On medium three and higher difficulty levels, when stability falls below roughly 30%, the entire canvas shakes. This effect is restricted to later, more difficult stages so that early levels remain learnable while advanced levels feel physically and visually destabilising.

**Decision 4: Mapping Focus Drain and Distraction Sound Cues into concrete systems**

New mechanics proposal for A3 included:

- Focus Drain: hesitation causing the board to darken or the timer to speed up.
- Distraction Sound Cues: unexpected sounds that draw attention away from the puzzle.

In the final game:

- Focus Drain is expressed indirectly via the stability/health bar and screen shake. As players take longer and stability drops, they enter the low-health, shaking state, which makes it harder to concentrate without literally blacking out the board.
- Distraction Sound Cues are partially realised through Matt's layered level music and distinct error SFX. These audio elements punctuate gameplay and can momentarily hijack attention, adding another channel of anxiety.

## Role-Based Process Evidence

**New Mechanic Research**

Created by: Frank Le – Research Lead / Mechanics Support

Used by: Karen Wu – Technical Director / Lead Developer

Frank revisited the three original A2 mechanics (Puzzle Matching, Time Pressure, Intrusive Pop-Ups) and formally extended them into a richer mechanic set aimed at supporting a multi-level difficulty curve. This included proposing new mechanics such as Moving Puzzle Pieces, Focus Drain, and Distraction Sound Cues, each explicitly mapped to aspects of Generalized Anxiety Disorder (e.g., loss of control, cognitive overload, sensory sensitivity). Karen used this document as a blueprint to decide which mechanics should become full systems in the final build, which should be indirectly represented (e.g., Focus Drain via the

health bar and shake), and which would remain conceptual influences due to scope or technical constraints.

**Game Audio & Visual Design**

Created by: Matt Spong – Visual & Audio Designer / UX & Experience Designer

Used by: Karen Wu – Technical Director / Lead Developer

Once the mechanic direction was set, Matt developed the A3 audio-visual language for the new multi-level structure. This work included sourcing and creating level-specific music tracks, varied error sound effects, and 16:9 background designs that would scale from "very easy" through to "unpassable/interpretive" difficulty. Matt also explored anxiety-driven motion concepts, such as hourglass-themed backgrounds and moving environmental elements, to support the sense of rising instability across levels. Karen then translated these assets and concepts into the p5.js implementation, integrating them with the mechanic decisions from Frank's research while tuning readability and performance.

**Final Code Implementation & Integration**

Created by: Karen Wu – Technical Director / Lead Developer

Used by: Matt Spong – Visual & Audio Designer / UX & Experience Designer; Frank Le – Research Lead / Mechanics Support

After the New Mechanic Research and Game Audio & Visual Design phases, Karen led the implementation of the A3 systems: building out multiple levels with distributed difficulty, adding and synchronizing the health bar with the timer, introducing screen shake on higher difficulties, and integrating Matt's audio-visual assets. Matt and Frank playtested these builds, using their earlier research and design work to evaluate whether the resulting experience accurately reflected the intended anxiety mechanics and progression.

## *GenAI Documentation*

**Date Used**: April 12, 2026

**Tool Disclosure**: Google Gemini

**Purpose of Use**: Debugging and clarifying sprite-sheet behaviour for the hourglass UI in the A3 build

**Summary of Interaction**: GenAI was consulted when the hourglass graphics were not behaving correctly in code. Instead of two clean hourglasses on the left and right sides, multiple "ghost" images and repetitions appeared across the screen. It responded by:

- Explaining that missing `background() ` in the `draw() ` loop can cause old frames to remain on screen and create trails or ghosting.
- Showing the 9-parameter version of `image() ` in p5.js (`image(img, dx, dy, dWidth, dHeight, sx, sy, sWidth, sHeight) `) and describing how to use it to crop one frame from a long sprite sheet.
- Warning that if sprite drawing is inside a loop (for grid cells, particles, or pop-ups), it can unintentionally draw the same sprite many times across the canvas.

**Human Decision Point(s)**:

- Kept our existing project structure and variable names, and only borrowed the idea of using the 9-parameter `image() ` call with explicit source rectangles.
- Manually added or confirmed `background() ` at the top of `draw() ` and verified that the hourglass drawing was moved out of any grid or loop logic so it would be drawn only twice per frame.
- Ignored suggestions that would have required restructuring large parts of the rendering pipeline or adding extra animation complexity, because that would have risked new bugs and increased scope.

**Integrity & Verification Note**: After applying GenAI-informed changes, Karen verified the code in two stages:

1. Isolated test: Karen created or used a minimal sketch focused only on loading the hourglass sprite sheet and drawing a single frame with `image() `. This confirmed that the cropping values (`sx `, `sy `, `sWidth `, `sHeight `) were correct and that no ghosting appeared when `background() ` was called each frame.
2. **Full game test:** Karen re-inserted the corrected drawing logic into the full A3 build and checked that:
    - Only two hourglasses appeared at the intended positions.
    - No new trails or ghost images appeared over time.
    - The changes did not break other systems (grid interaction, timer, pop-ups, or health bar).

Karen also visually compared the before/after behaviour to make sure that the hourglass UI felt clean and readable and did not interfere with the grid or HUD.

**Scope of GenAI Use**:

GenAI's contribution was limited to:

- Explaining general sprite-sheet rendering issues in p5.js.
- Suggesting the correct use of `background() ` and the 9-parameter `image() ` function.

GenAI did not:

- Design or implement the game's architecture, mechanics, or level progression.
- Write entire scripts (such as `game.js `, `grid.js `, `sketch.js `, or the health bar logic).
- Decide where or how many hourglasses to draw, or how they relate to the health bar and difficulty levels.

All structural code, mechanics (multi-level progression, health bar, screen shake, pop-ups), and integration of assets were written and tuned by the team, with GenAI only helping resolve specific rendering bugs

**Limitations or Misfires**: GenAI's examples assumed generic sprite sizes and layouts, so the hourglass first appeared stretched or misaligned and Karen had to correct the source rectangle values. It also suggested extra animation loops that Karen intentionally skipped this late in the project.

### *Summary of Process (Human + Tool)*

Gemini was used late in development, when a bug appeared with the new hourglass sprites. After the backgrounds and hourglass graphics were added, the sprites began "ghosting" and repeating across the screen instead of staying as two clean images on the left and right.

### *Decision Points & Trade-offs*

Karen decided to use GenAI only as a last-resort debugging tool, not as a co-designer. When the ghosting bug appeared, the options were to keep manually experimenting with p5.js or briefly consult Gemini to confirm what was going wrong. Karen chose to use Gemini once to speed up identifying the cause, but rejected its more complex animation suggestions to avoid large late-stage code changes.

### *Verification & Judgement*

Any Gemini-informed edits were tested in code before being accepted. Karen first verified that `background()` and the new `image()` parameters worked correctly in a minimal sketch, then checked the full game to ensure the hourglasses rendered properly and nothing else broke (grid, timer, health bar, pop-ups).

*Limitations, Dead Ends, or Open Questions*

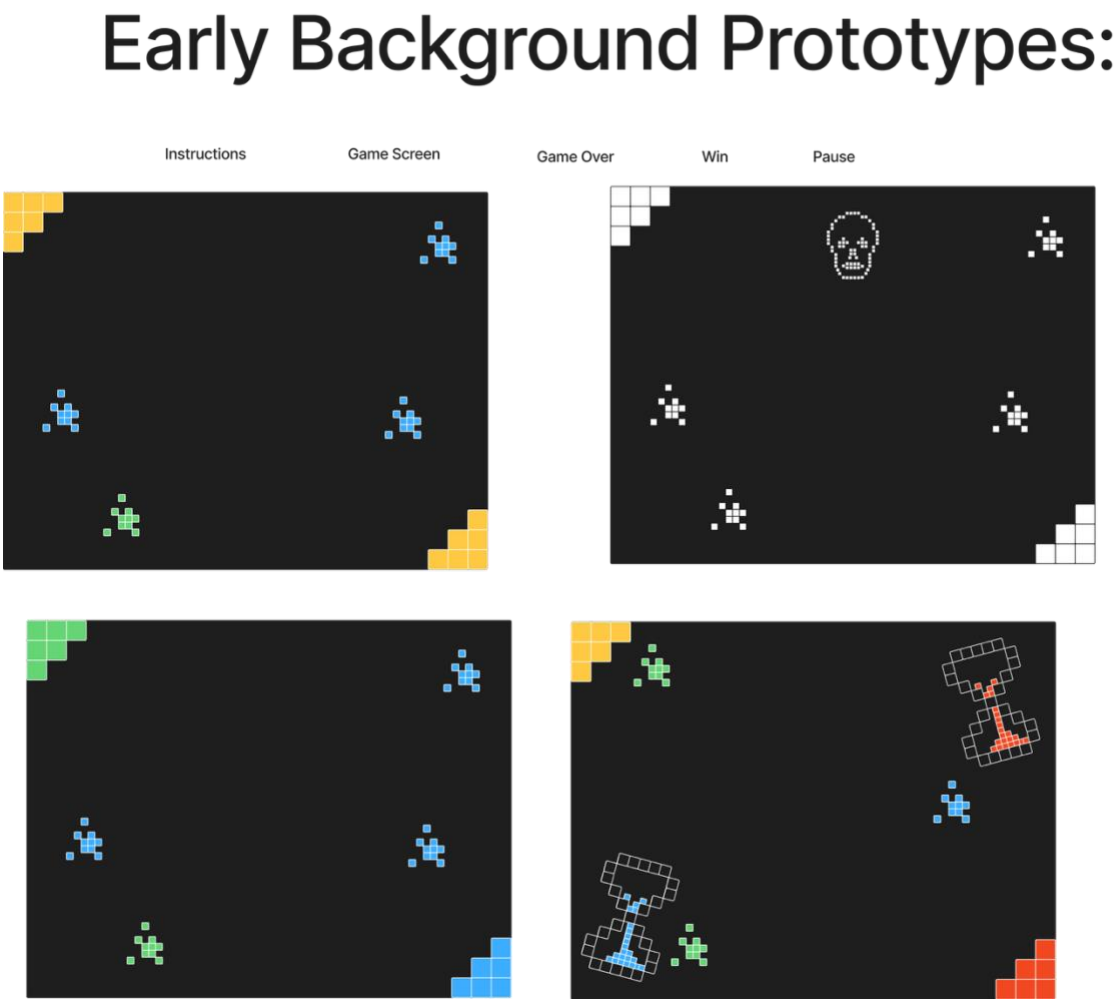
Gemini assumed generic sprite sizes and layouts, so its example values did not match the team's assets and initially caused stretching and misalignment. These had to be corrected manually.

## Appendix

**AI Transcript**

https://gemini.google.com/share/90d33bdc3a13

**Visual Assets**



Improved Game Backgrounds:
-Changed resolutions (16:9)
-Changed Colour Schemes
-Changed Hourglass sizes
-Added New Corner Elements

Final Background:

-Tilted Hourglass

-Made Hour Glass Smaller

-Changed 2 Page Colours